Template Screening Report (v5.7.1 - Multi-Source AI Evaluator)

Journal Template: Jurnal Teknologi

Submission No: 26804

Author Name: Kalnaowakul et al.

Article Title: RSM-ASSISTED ROSEMARY ESSENTIAL OIL PRECIPITATION FOR ELUCIDATING THE INTERPLAY OF NANOENCAPSULATION VIA RAPID HOMOGENIZATION DYNAMICS

Source File: 1. JT_RSM-assisted Rosemary Essential Oil Precipitation.doc

Screened PDF: 1. JT_RSM-assisted Rosemary Essential Oil Precipitation.pdf

Screening Date: 2026-05-21 09:43:07

Detection Source Debug Report

Item	DOC Detection	PDF Detection	AI Detection	Final Source
Article Title	NO	YES	YES	PDF + AI
Authors	NO	YES	YES	PDF + AI
Affiliations	NO	YES	YES	PDF + AI
Corresponding Author	YES	YES	SKIPPED	DOC + PDF
Graphical Abstract	YES	YES	SKIPPED	DOC + PDF
English Abstract	YES	YES	SKIPPED	DOC + PDF
Keywords	YES	YES	SKIPPED	DOC + PDF
Malay Abstract (Abstrak)	NO	YES	SKIPPED	PDF only
Kata Kunci	NO	NO	SKIPPED	None
Introduction	YES	NO	SKIPPED	DOC only
Methodology	YES	NO	SKIPPED	DOC only
Results and Discussion	YES	YES	SKIPPED	DOC + PDF
Conclusion	YES	NO	SKIPPED	DOC only
Acknowledgement	YES	YES	SKIPPED	DOC + PDF
References	YES	YES	SKIPPED	DOC + PDF

Summary: DOC detected 10 items | PDF detected 11 items | AI confirmed 3 items

The system uses DOC + PDF + AI voting logic. AI serves as a third checker when DOC and PDF conflict. AI is skipped when DOC and PDF already agree.

Executive Summary

Overall Decision: NEEDS MANUAL REVIEW

Main Reason: The manuscript generally follows the template but contains inconsistencies that require manual review.

Critical Failed Items: None

Items Needing Review:

• Article Title

- Authors
- Affiliations
- Malay Abstract (Abstrak)
- Introduction

Suggested Action: Editor should review the highlighted issues before proceeding.

Overall Result

Metric	Score / Value
Final Status	NEEDS MANUAL REVIEW
Final Score	76/100
Structure Score	47%
Layout Score	89%
Formatting Score	100%
Detection Method	DOCX-First + Multi-Source AI (v5.7.1)
Decision	MANUAL REVIEW

How Items Are Evaluated

DOC + PDF + AI SKIPPED	PASS	Both deterministic detectors agree
DOC + PDF + AI CONFIRMED	PASS	All three confirm
DOC + AI	NEEDS REVIEW	DOC+AI confirm, but PDF layout conflicts
PDF + AI	NEEDS REVIEW	PDF+AI confirm, but DOC source extraction fails
DOC only	NEEDS REVIEW	DOC detects, but PDF+AI don't confirm
PDF only	NEEDS REVIEW	PDF detects, but DOC+AI don't confirm
AI only	NEEDS REVIEW	Only AI confirms (deterministic detectors failed)
None	FAILED	All three fail to detect required item

Status Explanation

PASS — Item follows the template structure, position, and formatting.

WARNING/NEEDS REVIEW — Item exists but evidence is inconsistent or layout differs from template.

FAILED — Required item truly missing after DOC/PDF/AI checks.

Item Screening Details

Item	DOC	PDF	AI	Final Source	Status	Reason
Article Title	NO	YES	YES	PDF + AI	WARNING	Title detected by PDF and AI, but DOC source text did not co
Authors	NO	YES	YES	PDF + AI	WARNING	Authors detected by PDF and AI, but DOC source text did not
Affiliations	NO	YES	YES	PDF + AI	WARNING	Affiliations detected by PDF and AI, but DOC source text did
Corresponding Author	YES	YES	SKIPPED	DOC + PDF	PASS	Corresponding_Author confirmed by both DOC and PDF detection

Graphical Abstract	YES	YES	SKIPPED	DOC + PDF	PASS	Graphical_Abstract confirmed by both DOC and PDF detection.
English Abstract	YES	YES	SKIPPED	DOC + PDF	PASS	Abstract confirmed by both DOC and PDF detection.
Keywords	YES	YES	SKIPPED	DOC + PDF	PASS	Keywords confirmed by both DOC and PDF detection.
Malay Abstract (Abstrak)	NO	YES	SKIPPED	PDF only	WARNING	Abstrak detected in PDF layout, but DOC and AI did not confi
Kata Kunci	NO	NO	SKIPPED	None	FAILED	Kata_Kunci not found by DOC, PDF, or AI.
Introduction	YES	NO	SKIPPED	DOC only	WARNING	Introduction detected in DOC, but PDF and AI did not confirm
Methodology	YES	NO	SKIPPED	DOC only	WARNING	Methodology detected in DOC, but PDF and AI did not confirm.
Results and Discussion	YES	YES	SKIPPED	DOC + PDF	PASS	Results_Discussion confirmed by both DOC and PDF detection.
Conclusion	YES	NO	SKIPPED	DOC only	WARNING	Conclusion detected in DOC, but PDF and AI did not confirm.
Acknowledgement	YES	YES	SKIPPED	DOC + PDF	PASS	Acknowledgement confirmed by both DOC and PDF detection.
References	YES	YES	SKIPPED	DOC + PDF	PASS	References confirmed by both DOC and PDF detection.

Detailed Findings

Article Title [title]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Article Title' section must exist.

Found: Matched heading: "RSM-ASSISTED ROSEMARY ESSENTIAL OIL PRECIPITATION FOR ELUCIDATING THE INTERPLAY OF NANOENCAPSULATI" | Detection method: score_rank

Confidence: 1.00

Detection Method: score_ranking

DOC Detection: NO

PDF Detection: YES

AI Detection: YES

AI Confidence: 1.00

AI Evidence: PDF snippet shows a full title section with all expected components; DOC snippet is missing the title section entirely.

Evidence Sources: PDF + AI

Final Source Used: PDF + AI

Rejected false positives: Jurnal Teknologi, Full Paper, Article history

Status: WARNING

Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, top area

Found: Page 1, left column, upper area

Position Check: WARNING

Reason: Position deviates from expected region (tolerance: 0.10).

C. Formatting Evaluation

Expected: Heading should be in uppercase style.

Expected: Heading should use bold font.

Formatting Check: PASS

Reason: Formatting matches template expectations.

Overall Item Status: WARNING

Authors [authors]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Authors' section must exist.

Found: Matched heading: "Phuri Kalnaowakula, Chatchawan Sookmanb, Jittima Kalnaowakulc, Kanchana Luepongb*" | Detection method: score_ranking

Confidence: 0.90

Detection Method: score_ranking

DOC Detection: NO

PDF Detection: YES

AI Detection: YES

AI Confidence: 0.90

AI Evidence: PDF snippet shows full Authors section with names, affiliations, and corresponding author details. DOC snippet does not contain the Authors section, l

Evidence Sources: PDF + AI

Final Source Used: PDF + AI

Rejected false positives: Article history, Received, Accepted

Status: WARNING

Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, upper area

Found: Page 1, left column, upper area

Position Check: WARNING

Reason: Position deviates from expected region (tolerance: 0.10).

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.
Overall Item Status: WARNING

Affiliations [affiliations]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Affiliations' section must exist.
Found: Matched heading: "aFaculty of Industry and Technology, Rajamangala University of Technology Isan, Sakon Nakhon 47" | Detection method: score_rank
Confidence: 0.80
Detection Method: score_ranking
DOC Detection: NO
PDF Detection: YES
AI Detection: YES
AI Confidence: 0.80
AI Evidence: PDF snippet shows a clear Affiliations section with author names and affiliations. DOC snippet contains the same text but was not detected by the DOC
Evidence Sources: PDF + AI
Final Source Used: PDF + AI
Rejected false positives: Accepted
Status: WARNING
Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, upper area
Found: Page 1, left column, upper area
Position Check: PASS
Reason: Position matches expected template region.

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.
Overall Item Status: WARNING

Corresponding Author [corresponding_author]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Corresponding Author' section must exist.
Found: Matched heading: "Corresponding author" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: YES
AI Detection: SKIPPED
Evidence Sources: DOC + PDF
Final Source Used: DOC + PDF
Status: PASS
Reason: The required component 'Corresponding Author' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, upper area
Found: Page 1, right-center area, upper area
Position Check: WARNING
Reason: Position deviates from expected region (tolerance: 0.12).

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS

Reason: Formatting matches template expectations.

Overall Item Status: PASS

Graphical Abstract [graphical_abstract]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Graphical Abstract' component must exist.
Found: Matched heading: "Graphical abstract" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: YES
AI Detection: SKIPPED
Evidence Sources: DOC + PDF
Final Source Used: DOC + PDF
Status: PASS
Reason: The required component 'Graphical Abstract' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, upper area
Found: Page 1, left-center area, upper-middle area
Position Check: WARNING
Reason: Position deviates from expected region (tolerance: 0.12).

C. Formatting Evaluation

Expected: Image should be present.
Formatting Check: PASS
Reason: Formatting matches template expectations.

Overall Item Status: PASS

English Abstract [abstract]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'English Abstract' section must exist.
Found: Matched heading: "Abstract" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: YES
AI Detection: SKIPPED
Evidence Sources: DOC + PDF
Final Source Used: DOC + PDF
Status: PASS
Reason: The required component 'English Abstract' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, upper area
Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.

Overall Item Status: PASS

Keywords [keywords]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Keywords' section must exist.
Found: Matched heading: "Keywords" | Detection method: source_docx_text

Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: YES
AI Detection: SKIPPED
Evidence Sources: DOC + PDF
Final Source Used: DOC + PDF
Status: PASS
Reason: The required component 'Keywords' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left-center area, lower-middle area
Found: Page 1, left-center area, lower-middle area
Position Check: PASS
Reason: Position matches expected template region.

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.
Overall Item Status: PASS

Malay Abstract (Abstrak) [abstrak]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Malay Abstract (Abstrak)' section must exist.
Found: Nearby content: "Jurnal Teknologi Full Paper RSM-ASSISTED ROSEMARY ESSENTIAL OIL PRECIPITATION FOR ELUCIDATING THE INTERPLAY OF NANOENCAPSULATION VI
Confidence: 0.00
Detection Method: semantic_fallback
DOC Detection: NO
PDF Detection: YES
AI Detection: SKIPPED
Evidence Sources: PDF
Final Source Used: PDF only
Status: WARNING
Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, lower area
Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.
Overall Item Status: WARNING

Kata Kunci [kata_kunci]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Kata Kunci' section must exist.
Found: Detection method: not_found
Confidence: 0.00
Detection Method: not_found
DOC Detection: NO
PDF Detection: NO
AI Detection: SKIPPED
Evidence Sources: None
Final Source Used: None
Status: FAILED

Reason: Required component 'Kata Kunci' was not found.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, lower area
Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.

Overall Item Status: FAILED

Introduction [introduction]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Introduction' section must exist.
Found: Matched heading: "1.0 INTRODUCTION" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: NO
AI Detection: SKIPPED
Evidence Sources: DOC
Final Source Used: DOC only
Rejected false positives: 1.0 INTRODUCTION
Status: WARNING
Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area
Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.

Overall Item Status: WARNING

Methodology [methodology]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Methodology' section must exist.
Found: Matched heading: "2.0 METHODOLOGY" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: NO
AI Detection: SKIPPED
Evidence Sources: DOC
Final Source Used: DOC only
Rejected false positives: 2.0 METHODOLOGY
Status: WARNING
Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area
Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.

Overall Item Status: WARNING

Results and Discussion [results_discussion]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Results and Discussion' section must exist.
Found: Matched heading: "3.0 RESULTS AND DISCUSSION" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: YES
AI Detection: SKIPPED
Evidence Sources: DOC + PDF
Final Source Used: DOC + PDF
Status: PASS
Reason: The required component 'Results and Discussion' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area
Found: Page 1, center area, lower area
Position Check: WARNING
Reason: Position deviates from expected region (tolerance: 0.20).

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.

Overall Item Status: PASS

Conclusion [conclusion]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Conclusion' section must exist.
Found: Matched heading: "4.0 CONCLUSION" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: NO
AI Detection: SKIPPED
Evidence Sources: DOC
Final Source Used: DOC only
Rejected false positives: 4.0 CONCLUSION
Status: WARNING
Reason: Component found but evidence is inconsistent.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area
Found: Not detected

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.

Overall Item Status: WARNING

Acknowledgement [acknowledgement]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'Acknowledgement' component is recommended.
Found: Matched heading: "Acknowledgement" | Detection method: source_docx_text

Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: YES
AI Detection: SKIPPED
Evidence Sources: DOC + PDF
Final Source Used: DOC + PDF
Status: PASS
Reason: The required component 'Acknowledgement' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area
Found: Page 1, center area, top area
Position Check: WARNING
Reason: Position deviates from expected region (tolerance: 0.20).

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.

Overall Item Status: PASS

References [references]

A. Structure Evaluation (DOC + PDF + AI)

Expected: The 'References' section must exist.
Found: Matched heading: "References" | Detection method: source_docx_text
Confidence: 0.85
Detection Method: source_docx_text
DOC Detection: YES
PDF Detection: YES
AI Detection: SKIPPED
Evidence Sources: DOC + PDF
Final Source Used: DOC + PDF
Status: PASS
Reason: The required component 'References' exists.

B. Layout / Coordinate Evaluation

Expected: Page 1, left column, top area
Found: Page 1, center area, upper area
Position Check: WARNING
Reason: Position deviates from expected region (tolerance: 0.20).

C. Formatting Evaluation

Expected: Standard paragraph formatting.
Formatting Check: PASS
Reason: Formatting matches template expectations.

Overall Item Status: PASS

Editor Conclusion

The manuscript generally follows the template but has items that require manual review. These items had inconsistent evidence between DOC detection, PDF layout analysis, or AI verification. Please review the highlighted items before making a final decision.

Structure Score: 47% (Weight: 40%)

Layout Score: 89% (Weight: 30%)

Formatting Score: 100% (Weight: 30%)

Final Score: 76/100

Items Needing Review:

• Article Title — WARNING
• Authors — WARNING
• Affiliations — WARNING
• Malay Abstract (Abstrak) — WARNING
• Introduction — WARNING

This report was automatically generated by Turnitin Assistant v5.7.1 Template Screening Engine (DOCX-First + Multi-Source AI Evaluator).

Jurnal
Teknologi

Full Paper

RSM-assisted Rosemary Essential Oil Precipitation for Elucidating the Interplay of Nanoencapsulation via Rapid Homogenization Dynamics

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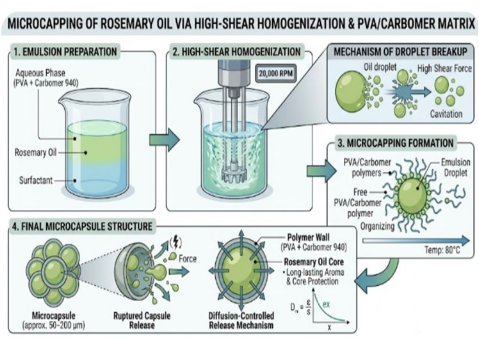
Graphical abstract



Abstract

Nanotechnology plays a pivotal role in various industrial and clinical applications, including food science, agriculture, and herbal medicine. This study aims to enhance the parameters for rosemary essential oil encapsulation by the solvent evaporation technique. These were used in response surface methodology (RSM) with a design to look at how changing variables like spinning speed, temperature, and concentration of rosemary essential oil (REO) affected the results. To achieve efficiency, adequate encapsulation investigation, such as Fourier transform infrared spectroscopy (FTIR), scanning electron microscopy (SEM), and Thermogravimetric analysis (TGA), was conducted. The effect of various homogenizer speeds on encapsulation efficiency (EE (%)), and REO were studied. The different of homogenizer speeds start from 1,000, 10,000 to 20,000 rpm combined with temperature and REO were presented the highest encapsulation efficiency (EE (%)). The maximum rotating speeds indicated that the EE (%) of REO encapsulated over 90%. SEM demonstrated that, among all components, the REO had the most pronounced spherical morphology. The FTIR spectra showed a suitable intramolecular bond between various parameters and REO. However, nanotechnology process can synergize to provide a robust foundation for industrial-scale production and further development of REO as a medicinal resource.

Keywords: Rosemary, Essential oil, Molecular distillation, Surfactant reagent assisted, homogenizer method.

1.0 INTRODUCTION

Nanotechnology is emerging as a primary influence in the advancement of the food, medical, and pharmaceutical industries. The application of nanoprecipitation composite process in this domain is wholly commendable. The main emphasis is on herbal formulations that enhance the efficacy and extraction of essential oil compounds through nanotechnology, thus enabling the amalgamation of traditional herbal therapy with technological innovations. In this context, the extraction of essential oils (EOs) has long been acknowledged as an intriguing method for the utilization of aromatic plants. Aromatic plant-based nanoparticles possess significant antidepressant and sleep-enhancing potential, as they can mitigate side effects, a frequent consequence of wellness interventions. Non-herbal formulations such as phytosomes, liposomes, capsules, and emulsions were examined to elucidate the mechanism of synergistic herbal essential oils and their application in nano-encapsulation [1, 2, 3]. The herbal industry has been substantially advanced by encapsulation technology, which has facilitated the incorporation and stabilization of aromatic active compounds within a variety of essential oil systems. This method entails the encapsulation of botanical active compounds within a carrier matrix, thereby enhancing their bioavailability, regulatory release, and stability. In order to optimize the health benefits of bioactive compounds, including vitamins, minerals, antioxidants, and probiotics, their effective distribution is necessary. Encapsulation not only protects these substances from harmful environmental conditions and interactions with other dietary constituents but also preserves their functional qualities [4, 5, 6]. The nutritional advantages of herbal active compounds are well-documented, with numerous studies underscoring their significance in health promotion, prevention, and therapy.

Rosemary (*Rosmarinus officinalis L.*) is a Mediterranean-native aromatic herbal therapy plant. Its potent medicinal properties, which include antioxidant, anti-inflammatory, antimicrobial, antidepressant, and sleep-enhancing effects, have garnered significant attention. Additionally, it functions as a seasoning in cosmetics, beverages, and culinary applications [7, 8, 9]. Rosmarinic acid, flavonoids, terpenoids, carnosic acid, and phenolic acids are the components of rosemary extracts (RE) that are responsible for their antioxidant and antibacterial properties. Rosmanol, rosmarinic acid, carnosol, carnosic acid, and tannin are the primary polyphenolic components present in the RE. The antioxidant and bioactive properties of rosemary are significantly improved by the phenolic components [10, 11, 12, 13]. Polyphenols are highly susceptible to heat and light, which can render them unstable. Additionally, they frequently do not dissolve readily in water, which complicates their utilization by the body. Additionally, the scope of their applicability in pharmaceutical and culinary settings is restricted. Microencapsulation has been identified as a viable approach to resolve these concerns, thereby enhancing the applicability of polyphenols in industrial settings, particularly in functional foods that are designed to provide health benefits [14, 15, 16]. Micro- or nano-encapsulation is a useful way to include delicate parts from rosemary extract into tiny bead forms, which helps make them more available and stable. The success of micro- or nano-encapsulation, how quickly the compounds are released, and the shape of the resulting micro- or nano-capsules (beads) are important features of the final product [17]. The high-speed homogenization technique can be employed to micro- or nano encapsulate the active constituent. Nevertheless, the lengths that were generated following the dispersion process remained micron-sized. Increased stirring velocities do not necessarily lead to significant changes in particle size; however, there is a slight increase in the polydispersity index. These nanoparticles are suitable for the incorporation of active substances, large-scale production, the improvement of product stability, and the omission of the need for organic solvents [18]. However, microencapsulation technologies pose certain obstacles. Elevated production expenses, intricate manufacturing processes, and scalability challenges are prevalent impediments. Moreover, interactions between encapsulated drugs and their surroundings might sometimes reduce efficacy. Resolving these difficulties is essential to enhance the technology and broaden its practical application across various sectors [19]. The development of more effective and efficient delivery systems is facilitated by the enhanced regulation of the release and targeting of bioactive substances that nanoencapsulation provides [20], [21]. These developments have the potential to significantly impact the future of food science, pharmaceuticals, and nutrition by facilitating the development of innovative products that meet the growing demand for health care solutions among consumers [22]. Consequently, the optimization of concentration parameters and data analysis are essential for the enhancement of manufacturing efficiency and the prevention of defects, as well as the extension of the life of encapsulation and the reduction of surfactant activity. In particular, online encapsulation monitoring systems are appealing for optimizing the encapsulation efficiency utilization ratio [23]. In addition to the information generated during the extraction process, a variety of statistical optimization techniques have been investigated and are available for the purpose of optimizing the input variables of absorbing processes. These techniques include Taguchi optimization, Grey relational analysis, Response Surface Methodology (RSM), Signal-to-Noise Ratio (SNR), and Analysis of Variance (ANOVA) [24]. RSM is a compilation of statistical and mathematical techniques that are used to model problems in which a response variable is needed to be optimized and is determined by multiple variables [25]. It has been employed to generate a mathematical model, optimize and analyze the experimental parameters, and provide significantly more information from a limited number of experiments. The utmost or minimum response values and the optimal factor settings (x) that produce the best response (Y) can be determined by examining response surfaces.

By overlaying the contours of the response surfaces of the regression models in a composite plot, the optimal conditions can be determined graphically. The RSM was extensively utilized in numerous research studies to examine the interactivity of independent variables across diverse domains, including analytical chemistry [26, 27, 28] and applied chemistry [29], [30]. Additionally, the stability and integrity of the encapsulated systems were elucidated through the investigation methods, including SEM, TGA, and FTIR, which were used to examine microstructures and morphologies. This study highlights practical applications, showcasing the potential of essential oil as a functional encapsulating medium for sensitive bioactive chemicals such as rosemary extract, advancing beyond previous research through systematic optimization and application-focused analysis.

2.0 METHODOLOGY

2.1 Materials

Carbomer 940, a low-viscosity material from Stongchemicals in Thailand, was used as the exterior material. The fundamental component was an essential oil, specifically Rosemary oil (Krungthepchemi, Thailand). The cross-linking material was composed of Polyvinyl alcohol (PVA, Sigma-Aldrich, USA) and Tween 80 (Chem-Supply, Australia).

2.2 Preparation of emulsion precipitation

The RSM method was created to dissolve PVA concentrations in distilled water, resulting in the preparation of carbomer and REO material solutions, as illustrated in Table 1. The solutions were maintained at ambient temperature overnight to calculate that the polymer molecules were completely hydrated. After that, Tween 80 served as emulsifier, mixed with REO in a 1:1 and 3:1 ratio. The REO solution was introduced at three speed data points for coating ratios of 1000, 10000, and 20000 rpm at 10 min. The REO solution was centrally located, with a layer of carbomer applied in varying amounts as described below.

Table 1 Carbomer and REO ratio prepared by RSM method

Group no.	Speed (rpm)	Temperature (C°)	REO (wt%)	Carbomer (wt%)
I	1,000	40	0.50	0.50
			2.25	0.75
			3.00	1.00
II	10,000	40	0.50	0.50
			2.25	0.75
			3.00	1.00
III	20,000	40	0.50	0.50
			2.25	0.75
			3.00	1.00
IV	1,000	60	0.50	0.50
			2.25	0.75
			3.00	1.00
V	10,000	60	0.50	0.50
			2.25	0.75
			3.00	1.00
VI	20,000	60	0.50	0.50
			2.25	0.75
			3.00	1.00
VII	1,000	80	0.50	0.50
			2.25	0.75
			3.00	1.00
VIII	10,000	80	0.50	0.50
			2.25	0.75
			3.00	1.00
VIIII	20,000	80	0.50	0.50
			2.25	0.75
			3.00	1.00

After this high-aggressively blending, the samples were examined by homogenizer (JSHR-270D, Korea). Subsequently, the sample particles were dried and placed in a humidity chamber as in Figure 1.

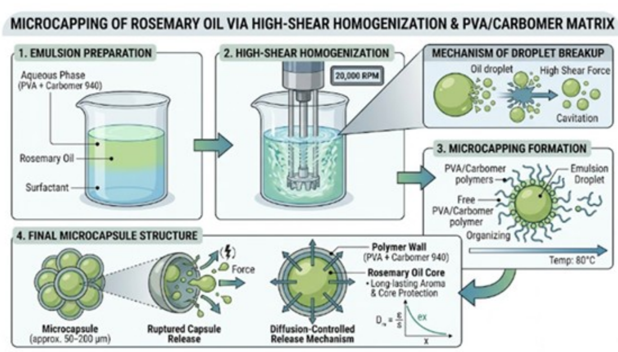


Figure 1 Preparation of REO nano encapsulate formation

2.3 Encapsulation efficiency

The EE (%) is determined to compare the amounts of distilled water constituents before and after encapsulation. 1 g of the microencapsulated substance was initially measured and combined with 100 mL of distilled water to serve as a solvent. Consequently, the UV-visible spectrophotometer (Libra S270, Biochrom USA) was employed to measure the total REO standard at λ_{max} (278 nm), as illustrated in Figure 2. In the same way, the surface REO content of microcapsules was determined by suspending nano-encapsules in distilled water for one minute and measuring the absorbance, as is approximately described in [31, 32, 33]. The following calculation formula was used to determine the EE (%) of nano-encapsules:

$$EE\ (\%) = \frac{\text{Initial REO concentration} - \text{Final REO concentration}}{\text{Initial REO concentration}} \times 100$$

The maximal EE (%) of nano-encapsules was obtained by all experiments that were repeated twice (97.44%), as illustrated in Table 2. The probable explanation is the effective utilization of blending parameters and the compatibility of complex polymers.

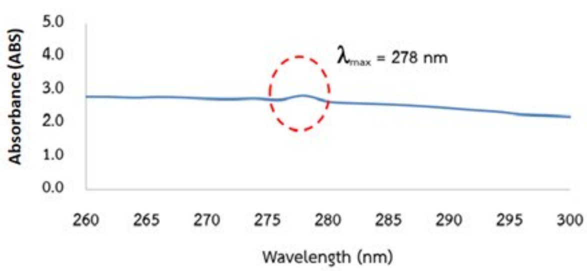


Figure 2 REO standard used by UV-visible photometer

Table 2 the maximum of emulsion concentration, encapsulation yield (EY %), and encapsulation efficiency (EE %) at 80 °C

Run no.	REO (wt%)	Carbomer (wt%)	PVA / Tween 80 (wt%)	EY (%)	EE (%)
1	0.5	0.5	0.25	65.26	82.02
2	0.5	0.5	0.25	65.60	82.24
		Average		**65.43**	**82.13**
3	2.25	0.75	0.25	70.15	89.42
4	2.25	0.75	0.25	70.27	89.92
		Average		**70.21**	**89.67**
5	3.00	1	0.25	76.87	97.41
6	3.00	1	0.25	77.43	97.47
		Average		**77.15**	**97.44**

2.4 Fourier-transformed infrared spectroscopy (FTIR) analysis

The Spectrum FTIR spectrometer (PerkinElmer, USA) was used to analyze the functional groups and provide information on the structural properties of the samples. At a scanning resolution of 1 cm^{-1}, the wavelengths of all spectra are recorded between 4000 and 400 cm^{-1}.

2.5 Thermogravimetric analysis (TGA) details

The thermal stability film of the active oil was analyzed using the TGA instrument (TGA/DSC 3 Plus, Mettler Toledo, Switzerland). Under a nitrogen purge gas flow rate of 20 mL/min, samples weighing 5 mg were heated at a rate of 10°C/min within a temperature range of 25 to 200°C.

2.6 Scanning electron microscopy (SEM) setup

The active films (1 cm × 1 cm) were affixed to a carbon tape on an aluminum holder, and a gold layer was sputter-coated on the top surface. The JSM-IT700HR (JEOL, Japan) was used to observe the surface and cross-sectional micrographs of the active films at an operating voltage of 5.00 kV.

2.7 Statistical data

The functional qualities of affected REO following duplicate, utilizing Minitab software (Minitab, Inc., U.S.A.). All statistical analyses were conducted with a 95% confidence level of significance; p-values less than 0.05 indicate a substantial difference in the data. In order to assess the variability of the data points that were acquired, descriptive and fixed statistics were calculated. The Response Surface Methodology (RSM) was employed to investigate the influence of process parameters on encapsulation efficiency (EE%). Statistical significance was evaluated using Analysis of Variance (ANOVA), where the null hypothesis (H_0) postulated that there were no significant factor effects on encapsulation efficiency. In contrast, the alternative hypothesis (H_1) proposed that there were significant factor influences.

2.8 pH setup

The samples were placed in a controlled environment (25°C ± 1°C, 72% ± 12% relative humidity) for 1 h., to assess the pH. The samples were aseptically transferred to Petri plates and placed in an incubator. The pH levels of the samples were re-evaluated after 1, 3, 5, 7, 9, 11, and 13 days of environmental exposure. For each sample, we measured 0.5 weight percent (wt.%) of REO and diluted it in 10 mL of distilled water. The pH was assessed thrice post-homogenization with a pH meter.

3.0 RESULTS AND DISCUSSION

3.1 The chemical makeup of rosemary essential oil (REO)

The results of the chemical composition analysis of REO are summarized in Table 3. In accordance with Table 3, the primary components of REO are camphor, 1,8-cineole, 3-carene, α-pinene, trans-caryophyllene, and β-pinene. These findings align with established knowledge of the primary chemical constituents of REO as documented in the literature (Ahsaei *et al*., 2020 [34]; Yeddes *et al*., 2020 [35]).

Table 3 Chemical constituents of the rosemary essential oil (REO)

Components	Percentage (%)
β-Bisabolene, β-Caryophyllene	0.30
Lavandula, 3-Keto-isosteviol	1.13
Fenchone	1.30
Aromadendrane, Viridiflorol, α-Santalol	1.80
Acetaldehyde, Camphor, β-pinene	2.48
δ-Cadinene	3.40
Cymene, α-pinene	3.84
Terpineol, Borneol	4.31
Trans-caryophyllene, Isoborneol	5.54
γ-Cadinene, Copaene, α-ylangene, Cycloisosativene	13.97
β-Propiolactone	14.44
α-Terpinolene, Camphene, 3-Carene	16.50
1,8-Cineole	30.91

3.2 SEM of REO encapsulated results

The morphology of the REO microencapsulated was analyzed using SEM images, as shown in Figure 3. The REO encapsules exhibit spherical, oval, and irregular surfaces, which are typical of homogenization encapsules. Reservoirs and matrices constituted the manifestation of the homogenizer encapsulates. The morphology of the encapsulate can be affected by the polymer type employed; nonetheless, spray-dried microcapsules attain equivalent outcomes (Zhu *et al*., 2021 [36]). During emulsion formation, the crosslinking interactions between Carbomer 940 and Tween 80 polymers established a continuous covering on the oil phase.

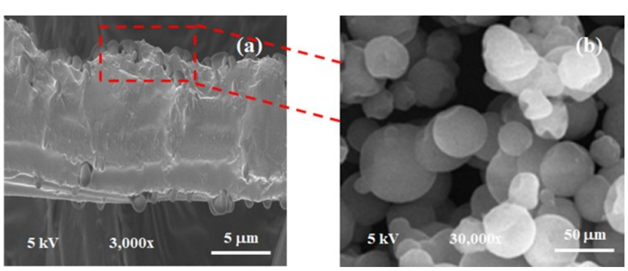


Figure 3 SEM of REO encapsulated at different magnifications: (a) 3,000x and (b) 30,000x

3.3 FTIR spectroscopy results

The spectrum of REO is shown in Figure 4. The molecular-level variation of chemical groups in composites is frequently investigated using FTIR analysis. Terpinolene, 1,8-cineole, and camphene comprise the majority of the components. Several components comprise the overall content, with pinene isomers serving as the primary component. Each of these components contributes to the C–H stretching bands at 1430 cm^{-1} and 3285-2865 cm^{-1} (in the functional group region). The ether functionality of the aliphatic polyether epoxy ring of 1,8-cineole is the component for the peaks at 1150 and 927 cm^{-1}. Additionally, the peak at 1028 cm^{-1} is obtained with the asymmetric elongation of the C-O bond [37]. Nevertheless, the two spectra exhibited some discrepancies, including a minor absorption band at approximately 850-400 cm^{-1} and a pronounced peak at wavenumbers 1250-1000 cm^{-1}. They are linked to the intermolecular bonds of the functional groups of polysaccharides. These findings are in agreement with the findings of Fatemeh and Saeed *et al*., 2019 [38].

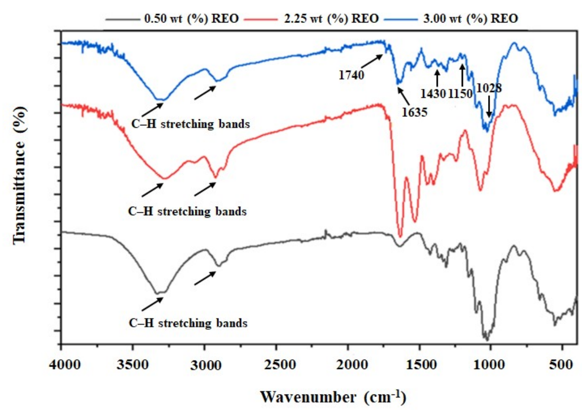


Figure 4 FTIR spectra of REO encapsulated film

3.4 TGA analysis results

The REO encapsule films during heating, TGA, were implemented. Figure 5 illustrates that the weight loss for all films followed a consistent pattern during the heating process, which consisted of two phases. The initial degradation occurred within a temperature range of 30–100 °C, and it was primarily caused by the vaporization of free and loosely bound water in the film. The secondary degradation, which potentially corresponded to the thermal decomposition of the polymer molecular chains and the release of the essential oils, took place between 100 and 200°C. The material undergoes carbonization as its ultimate degradation (Ji *et al*., 2022 [39]). The thermal stability of the films is improved by the addition of 3 wt% REO composite, as evidenced by the lower mass losses in the second and final stages compared to 0.5 wt% and 2.25 wt% of REO film. This may be attributed to the interaction between REO and polymer, which enables the formation of a more compact network structure within the films (Liu *et al*., 2023 [40]). In addition, the initial phase commences with the moisture in the exterior component at 100˚C, whilst the subsequent phase is triggered by the water within the encapsulates at temperatures ranging from 100˚C to 200˚C. The initial phase of the nanocomposite film exhibited less mass loss from water evaporation, attributed to the lowered moisture content, as reported by Pan *et al*. (2024) [41]. The decomposition of the essential oil at elevated temperatures is the primary factor influencing mass loss due to the films that contained essential oils exhibiting a reduced water content in this investigation.

Consequently, the TGA profiles underwent substantial modifications during mass loss.

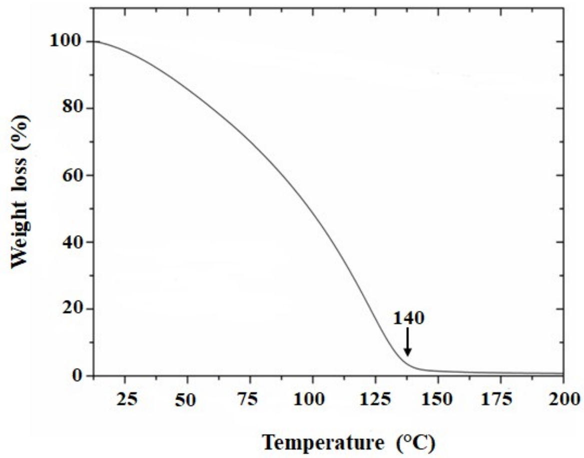


Figure 5 TGA diagram of REO encapsulates

3.5 Response surface optimization of the REO results

The response surface test results were used to derive the final model equation, which allowed for item-by-item testing. The four factors of speed (A), temperature (B), REO (%) (C), and carbomer 940 (%) (D) were fitted with a polynomial to establish a multivariate quadratic regression equation. The equation is as follows: EE (%) = 37.58 + 0.0022A + 10.10C – 0.0005AC. The results indicated that the corresponding T and P-values were used to evaluate the multiple regression analysis and the quadratic model of the response surface for EE (%), as presented in Table 4.

Table 4 RSM effect of the rosemary essential oil (REO)

Term	T-Value	P-Value
A	3.18	0.006
B	0.07	0.943
C	2.61	0.019
D	-0.49	0.634
A*A	0.16	0.876
B*B	0.75	0.465
C*C	0.53	0.601
D*D	-0.45	0.66
A*B	1.52	0.148
A*C	-2.44	0.027
A*D	0.84	0.415
B*C	-0.88	0.391
B*D	-1.44	0.168
C*D	-0.19	0.85

The ANOVA results presented in Table 4 revealed that among the four investigated factors, only two demonstrated statistically significant effects on encapsulation efficiency at $\alpha = 0.05$:

- Rotational speed (Factor A): T-value = 3.18, $p = 0.006$ ($p < 0.01$)
- REO concentration (Factor C): T-value = 2.61, $p = 0.019$ ($p < 0.05$)

Conversely, Factor B (T-value = 0.07, $p = 0.943$) and Factor D (T-value = -0.49, $p = 0.634$) exhibited no statistically significant influence on the response variable, indicating that these parameters do not substantially affect encapsulation efficiency within the experimental range studied. For the interaction terms evaluated, only the A×C interaction (rotational speed × REO concentration) demonstrated statistical significance (T-value = -2.44, $p = 0.027$). This negative T-value suggests an antagonistic interaction between these factors, indicating that the combined effect of increased rotational speed and REO concentration is less than the sum of their individual effects. Statistical significance was not achieved for any of the other two-factor interactions ($p > 0.05$). Figure 6 illustrates that the relationship between significant factors and encapsulation efficiency is predominantly linear within the experimental design space, as indicated by the non-significance of the quadratic terms (A×A, B×B, C×C, D×D).

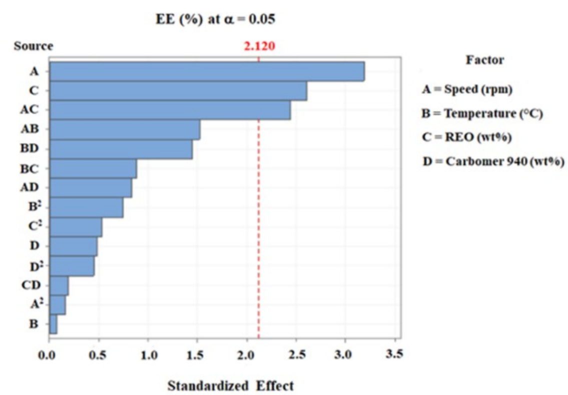


Figure 6 Factor of EE (%) for REO encapsulation

Model adequacy was rigorously validated through residual analysis (Figure 7), encompassing:

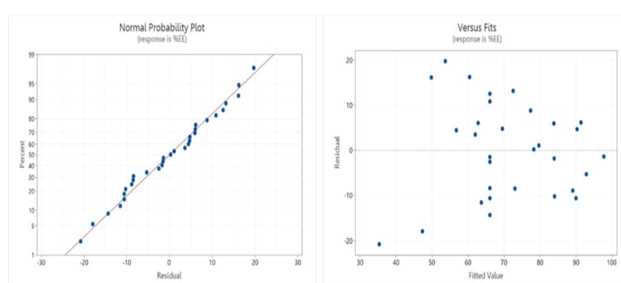


Figure 7 The RSM model supports the reliability of the statistical conclusions from the analysis

Residuals versus Fitted Values Plot:

(1) Normal Probability Plot (Q-Q Plot): The approximately linear arrangement of residuals along the theoretical normal line confirms that the residuals follow a normal distribution, satisfying the normality assumption required for valid statistical inference.

(2) Residuals versus Fitted Values Plot: The random scatter of residuals around zero with constant variance across the range of fitted values indicates

homoscedasticity, confirming that the model adequately captures the systematic variation in the data without detectable bias.

3.6 pH analysis results

The pH values exhibited minimal change, as seen in Figure 8. To ascertain the balanced activity of REO emulsions, we assessed them at different concentrations. The presence of REO did not influence pH readings at different concentrations during the 13-day study period. The initial pH of the REO emulsions was around 5.85, as observed. In a previous study by Qin *et al*., 2024 [42], a trend was observed that initially declined and subsequently rose as the duration of storage increased. The first reduction in pH results from the breakdown of ATP within the REO shell, which increases the concentrations of carbomer 940 and Tween 80. In accordance with Ran *et al*., 2024 [43], the pH was elevated. The pH levels of the different REO groups had significantly increased by the conclusion of the five-day incubation period, reaching a value of 6. Nevertheless, the pH value of the 5.85 group exhibited the slightest average increase trend, which was 5.60 on the 14th day ($P < 0.05$). These findings suggest that the REO film exhibited efficient preservation capabilities by preventing the formation of evaporation processes.

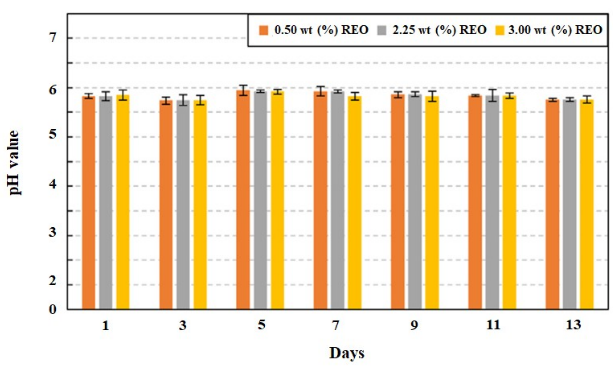


Figure 8 pH test of REO samples on different days

4.0 CONCLUSION

This study successfully fabricated REO encapsulates and integrated with a polymer matrix. SEM analysis confirmed that the film exhibited spherical, oval, and irregular surfaces, while pH test showed that the addition of composite REO did not alter the pH values at various concentrations. TGA and FTIR release, corroborate that homogenizer encapsulation process is optimal at 80˚C and it does not cause loss of oil. The concentration of the cross-linking agents, Carbomer 940 and Tween 80, is essential for the creation of the film or shell. The use of a solution and homogenization resulted in somewhat large encapsulations. The intended use dictates the suitability of REO encapsulates. The identification of REO concentration as the sole significant factor, along with its antagonistic interaction, provides critical insights for process optimization. The non-significance of quadratic effects suggests that optimization can be effectively achieved through linear modelling approaches within the studied experimental domain, potentially simplifying subsequent optimization procedures. The R^2 values were not reported as this investigation focused solely on factor significance determination rather than predictive model development for response optimization.

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